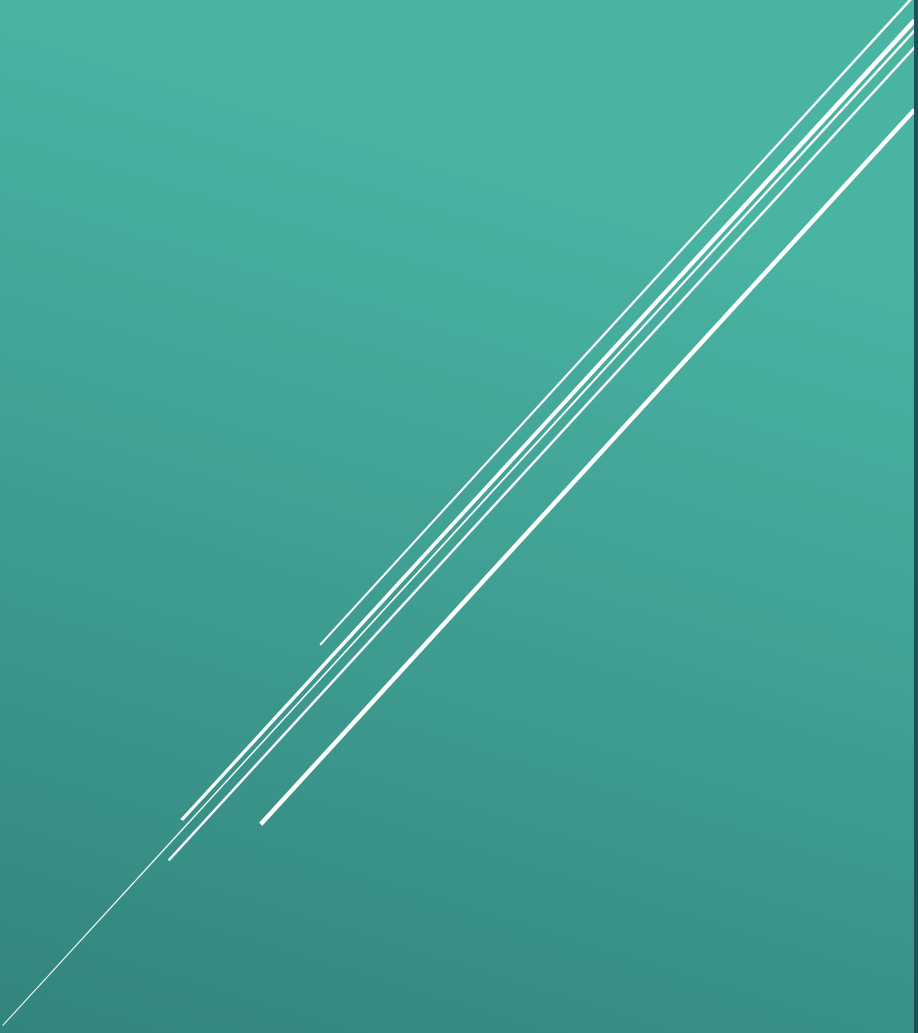




ASSESSMENT -1

Hardware Components & System Assembly

Section A - Concept Application

1. **SCENARIO:** You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

Question: Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

To choose hardware for a new office computer:

- First, check the ****CPU socket****.
- Select a motherboard with the ****same CPU socket****.
- Check that the motherboard supports the correct ****RAM type****, such as DDR4 or DDR5.
- Check that the motherboard supports the storage type, such as ****SATA or M.2 NVMe****.
- Check the motherboard manufacturer's specifications before buying.

2. **SCENARIO :** You are advising a client whose employees mainly run office and productivity applications on their workstations.

Question: Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

There are three common types of storage:

- ****HDD:**** Uses spinning disks. It is cheaper but slower.
- ****SSD:**** Uses flash memory. It is faster and quieter than HDD.
- ****NVMe SSD:**** Uses PCIe and is generally faster than a SATA SSD.

- ****For an office computer, I would choose an NVMe SSD**** because it gives fast booting, fast application loading and quick file access.

3. SCENARIO :You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

- The ****GPU**** processes graphics and helps design software run smoothly.
- The ****cooling system**** removes heat from the CPU and GPU.
- When the computer runs design software for many hours, it produces more heat. Good cooling prevents ****overheating and performance reduction****.

4. You are assembling a PC from components sourced from three different vendors.

Question: Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

- **Hardware compatibility means that all computer components can work together correctly.**

If components are incompatible:

- The computer may not start.
- RAM may not work.
- CPU may not fit the motherboard.
- Storage may not be detected.
- PSU may not provide enough power.
- Components may not work correctly.

5. You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

Question: Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

- I choose a PSU with enough ****wattage**** for the CPU, GPU, motherboard and other components.

Before turning on the PC, I check:

- 24-pin motherboard power is connected.
- CPU power cable is connected.
- GPU power cable is connected if required.
- Storage power cables are connected.
- All components are properly seated.
- No loose screws are inside the case.
- Power is disconnected while installing components.
- ESD precautions are followed.

6. You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

- POST means ****Power-On Self-Test****.

When I turn on the PC, I check:

- Does the PC power on?
- Do the fans start?
- Is there a display?
- Does BIOS open?

- Is the CPU detected?
- Is the correct RAM detected?
- Is the storage detected?
- Are there any beep codes or warning lights?

➤ **If POST fails:**

I would:

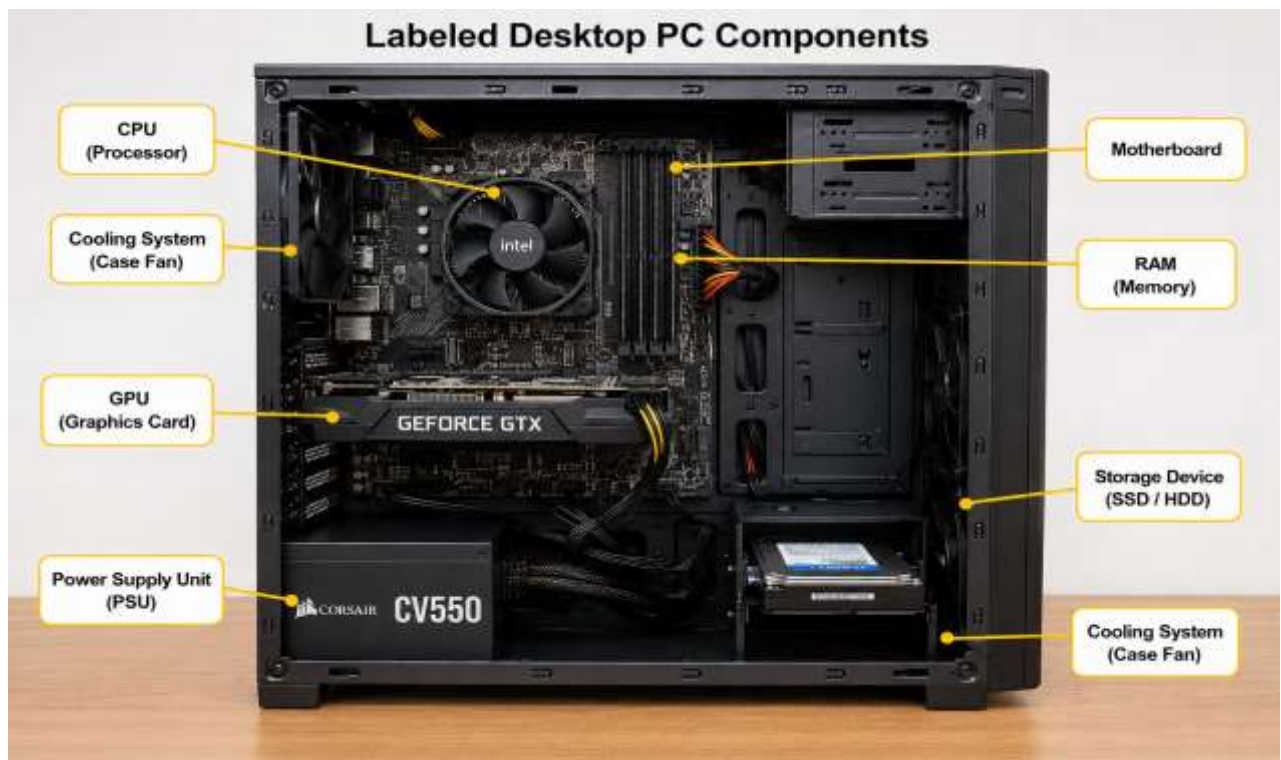
- Turn off the PC.
- Check PSU cables.
- Check RAM.
- Check CPU and CPU power.
- Check GPU.
- Check motherboard error lights/beeps.
- Check the motherboard manual.

Section B - Practical Tasks

TASK-1 Component Identification & Functions

Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system.

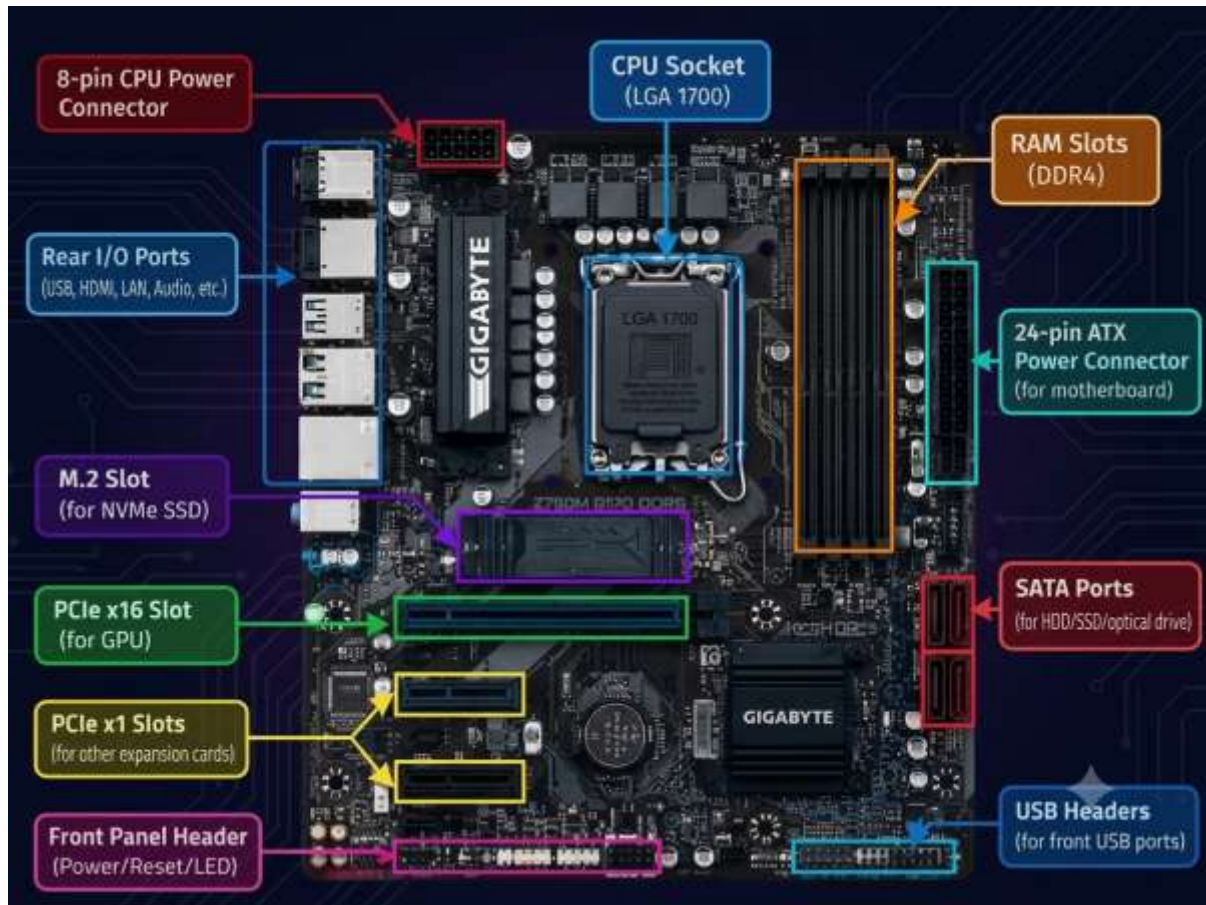
Note the make/model or key specification of each component where visible. Present the findings as a labeled component list or table.



Component	Function
Motherboard	Connects all components and allows them to communicate
CPU	Processes instructions and performs calculations
RAM	Temporarily stores data and programs currently in use
GPU	Processes and renders graphics and video
Storage	Stores files, applications, and the operating system
Cooling System	Removes heat and keeps components at a safe temperature

TASK-2 Motherboard Ports & Socket Identification

Identify and label all major motherboard ports (SATA, power, USB headers, etc.). Identify all expansion slots (PCIe, M.2, etc.) and their purpose. Identify the CPU socket type and note which CPU families are compatible with it. Photograph or diagram the motherboard with labels.



1. Major Ports & Headers

- **SATA Ports:** Connect HDDs and SSDs.
- **24-pin ATX Power:** Main motherboard power supply.
- **8-pin CPU Power:** Dedicated processor power.
- **Rear I/O Ports:** Back plugs for USB, HDMI, LAN, and audio.
- **USB & Front Panel Headers:** Connect case buttons, LEDs, and front USB ports.

2. Expansion Slots

- **PCIe x16 Slot:** For Graphics Cards (GPU).
- **PCIe x1 Slots:** For extra Wi-Fi or sound cards.

- M.2 Slot: For fast NVMe SSD storage.
- ### 3. CPU Socket & Compatibility
- Socket: LGA 1700
 - Compatible CPUs: Intel 12th, 13th, and 14th Gen Core Processors.

TASK-3 RAM & Storage Installation

Safely remove the RAM module, noting ESD precautions taken. Reinstall the RAM module correctly and confirm it is seated properly. Remove and reinstall the storage device, noting the connection type used (SATA/NVMe). Confirm both components are detected correctly after reinstallation.

❖ Remove and Install RAM



Step 1: Remove RAM

1. **Safety:** Wear an ESD wrist strap to prevent static damage.
2. **Unlock:** Push down the plastic tabs at the ends of the RAM slot.
3. **Remove:** Pull the RAM stick straight up.

Step 2: Reinstall RAM

1. **Align:** Match the notch on the bottom of the RAM stick with the slot divider.
2. **Press:** Push down firmly on both ends until you hear a "**CLICK**".
3. **Confirm:** Check that the side tabs are locked into place automatically.

❖ Remove and install SATA SSD



1. How to Connect SATA SSD

- **Align Notch:** Align the SATA SSD connector pins with the slot on the motherboard.
- **Push & Connect:** Push the SSD firmly into place until it clicks securely.

2. How to Remove SATA SSD

- **Disconnect Cables:** Unplug the SATA data and power cables connected to the SSD.
- **Detach SSD:** Gently pull and detach the SSD safely from the board.

- ✓ Start the computer Check BIOS/Windows to confirm that the RAM and storage device are detected correctly.

Task 4: Full Desktop PC Assembly

Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU. Follow correct installation sequence and ESD/safety precautions throughout. Document each step with labeled photos or diagrams. Power on the system and confirm it passes POST successfully



Step to Assemble PC

- **CPU Installation:** Carefully place the processor (CPU) into the socket on the motherboard.
- **RAM Installation:** Push the memory sticks (RAM) into the slots .
- **M.2 SSD (Storage):** Insert the small M.2 drive and secure it with a tiny screw.
- **SATA SSD (Storage):** Connect the larger 2.5-inch drive using data and power cables.
- **Motherboard into Case:** Lay the motherboard inside the computer case and screw it down.
- **PSU Installation:** Place the power supply (PSU) in the case and screw it in place.
- **Power Connections:** Connect all power cables from the PSU to the motherboard and components.
- **System Assembly:** Install the graphics card (GPU) and complete the final build.

Section C — Mini Capstone Project

Mini Project: Workstation Deployment & Boot-Failure Diagnosis

➤ SCENARIO :

ABC Solutions is setting up a new computer lab. After assembling several systems, some computers fail to start while others display hardware detection errors, and the IT team suspects incorrect installation or compatibility issues. Build (or inspect) a workstation, diagnose the likely causes of these failures, and produce a diagnostic and installation-verification report — combining component selection, installation procedure, and compatibility troubleshooting from Section B.

Ans :-

1. Three Hardware Components to Check First

1. SMPS/Power Supply – Check power cables and 24-pin/CPU power connectors.
2. RAM – Check whether RAM is properly inserted and compatible.
3. CPU – Check CPU installation, CPU power cable, and compatibility.

2. Motherboard–CPU Compatibility

- The CPU must be compatible with the motherboard's socket, chipset, and BIOS.
- If they are not compatible, the PC may not boot or may show a CPU error during POST.

3. RAM and Storage Verification

- RAM: Make sure it is properly locked in the slot. BIOS should show the correct RAM size.
- Storage: Check SATA cables or M.2 installation. BIOS should detect the SSD/HDD as a storage device.
- POST: It should complete without hardware errors.

4. Hardware Installation Checklist

- Install CPU and cooler correctly.
- Insert RAM properly until the clips lock.
- Connect motherboard and CPU power cables.
- Connect SSD/storage correctly.
- Check BIOS/POST for hardware detection.

5. Why SSD Instead of HDD?

SSDs are better for modern office PCs because they provide:

- Faster boot time
- Faster application loading
- No moving parts
- Less noise
- Better overall performance

✓ **Lab Use:** SSDs will make ABC Solutions' office computers faster and more responsive.

Section D — AI-Augmented Learning

STEP 1 · BUILD WITH AI

- Use an AI tool of your choice (ChatGPT, Claude, Copilot, etc.) to help you produce a step-by-step troubleshooting checklist that:

Step 1: Exact prompt given to AI

- Create a simple-English, step-by-step troubleshooting checklist for a junior technician diagnosing a desktop PC that fails to boot immediately after assembly. Put the checks in the correct priority order. Include ESD and electrical safety, PSU and power connections, motherboard and CPU, RAM, GPU and display, storage, BIOS/POST, and the minimum-hardware test. Apply the checklist to the ABC Solutions computer-lab scenario, where some newly assembled computers fail to start and others show hardware-detection errors. Explain which step would identify each likely problem. Do not assume exact component models.

❖ AI's original checklist

- Check that the monitor and keyboard are connected.
- Turn on the PSU and press the computer's power button.
- Check that the PSU has enough wattage.
- Check that the motherboard is compatible with the CPU.
- Check that the RAM is installed correctly.
- Check that the storage device is connected.
- Check the GPU and its power cable.
- Check BIOS settings and error messages.
- Replace components that may be faulty.
- Reinstall the operating system if the PC still does not boot.

❖ My corrected checklist

➤ Corrected Checklist

1. Turn off the PC, unplug power and follow ESD safety.
2. Check the PSU and all power cables.
3. Check the motherboard and CPU installation.
4. Check CPU and motherboard compatibility.
5. Remove and reseat the RAM.
6. Check the GPU and its power cable.
7. Check the storage connection.
8. Turn on the PC and check POST/BIOS.

9. Check motherboard error lights or beep codes.
10. Test components one at a time if the problem continues.

STEP 2 · TEST & DEBUG

What I Changed

I added ESD safety as the first step because the AI checklist did not clearly mention it. I also added CPU and motherboard compatibility checking. These checks should be done before testing software because the problem occurred immediately after assembly. The corrected checklist is therefore safer and more suitable for a technician.